

CONFERENCE PAPER PRESENTATION · GREEN AI

A Comparative Analysis of Energy Efficiency Across Large Language Models

Route each prompt to the smallest capable model

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Outline

1 Introduction

2 Motivation

3 Related Works & Gap Analysis

4 Problem Definition

5 Proposed Methodology

6 Architecture

7 Flow Charts

8 Algorithm

9 Implementation

10 Performance Evaluation

11 Limitations & Future Works

12 Conclusion

13 References

Introduction

Background

- LLMs (GPT, Llama, Mistral) handle billions of requests daily
- Most serving systems send every prompt to the same large model
- Prior work covers energy measurement, quantization, and routing, but separately
- Energy is rarely measured directly from GPU power in routing studies

Our Contributions

1

Profile

Energy study across model sizes and quantization levels using real-time NVML power.

2

Route

Lightweight prompt-complexity router choosing the smallest capable quantized model.

3

Evaluate

Adaptive routing vs. static large model on energy, quality, and latency.

Motivation

1

Usage is exploding

LLMs now serve billions of requests, from simple lookups to multi-step reasoning.

2

Uniform serving wastes energy

Every prompt goes to the same large model, whatever its difficulty.

3

Environmental cost is real

Large NLP models carry carbon and water footprints that are hard to ignore.

TODAY'S SERVING

Sentiment query

Multi-step math

Same large model

The easy prompt pays the same energy bill as the hard one.

Related Works & Gap Analysis

Measurement

Samsi et al. [3], TokenPowerBench [7]

Reliable energy numbers, but no serving-time decision.

Quantization

GPTQ [8], SmoothQuant [10], LLM.int8() [11], AWQ [16]

Compresses one model; no per-prompt choice among models.

Routing / cascading

FrugalGPT [12], RouteLLM [13], Hybrid LLM [14]

Optimizes money or quality; energy is estimated, not measured.

THE GAP

No prior work (1) measures energy from real-time GPU power, (2) treats quantization level as a routing variable, and (3) picks the model per prompt by complexity.

Problem Definition

Input

A user prompt q of unknown difficulty and a pool of quantized models in 3 tiers (1B, 3B/3.8B, 7-8B) within 8 GB VRAM.

Objective

Pick the smallest capable tier for q so that net inference energy E_{net} (Joules) is minimized.

Constraints

Keep quality near the Tier 3 baseline ($\text{QPR} \approx 100\%$), keep latency low, and keep router overhead negligible.

Key risk: routing a hard prompt to a small model gives a failed answer, so uncertain prompts must escalate to a larger tier.

Proposed Methodology

1

Tiered model pool

Quantized GGUF models grouped into 3 tiers under an 8 GB VRAM budget.

2

Complexity router

CPU-side features (heuristics + MiniLM) feed a classifier that picks the tier in under 5 ms.

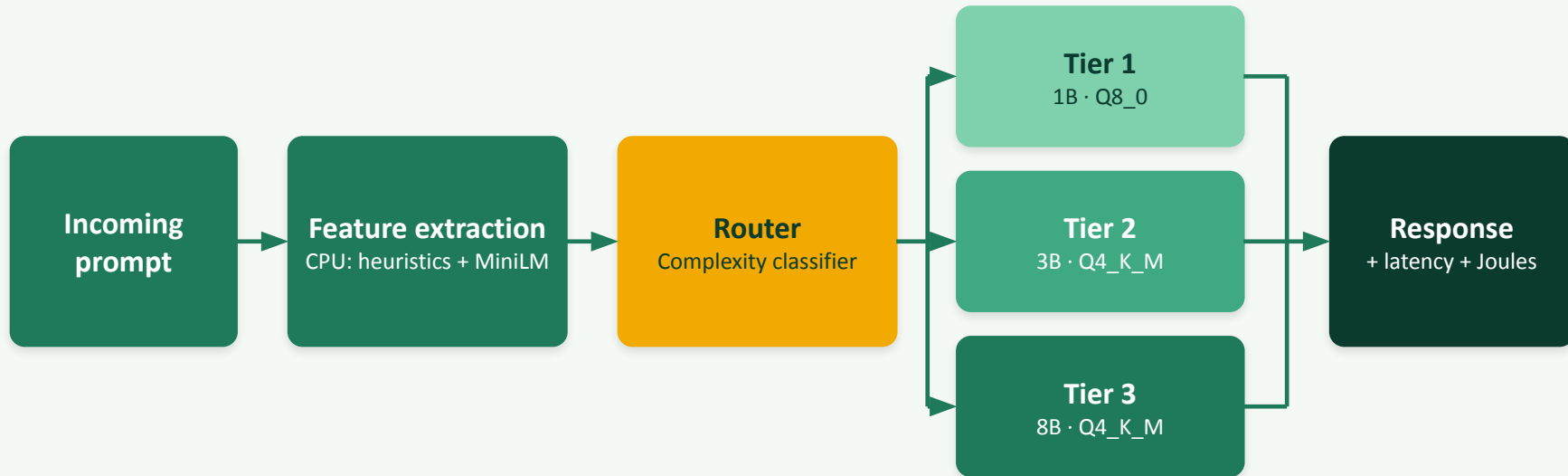
3

Energy profiling

NVML samples GPU power at 20 Hz; trapezoidal integration gives Joules per request.

Evaluated against a static Tier 3 baseline on energy, quality (QPR), and latency.

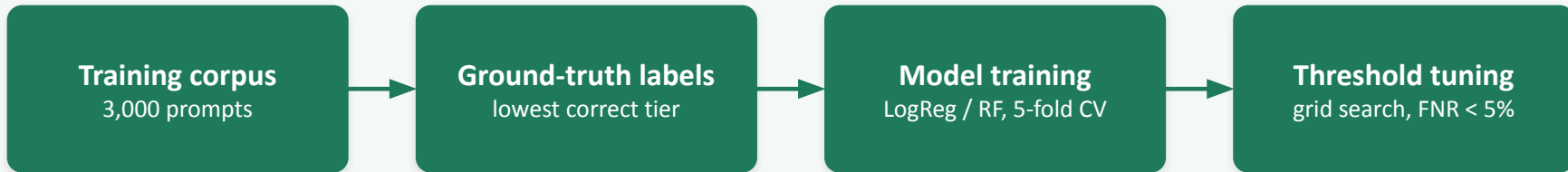
Architecture



Energy profiler (pynvml daemon, 20 Hz): samples GPU power $P(t)$ during generation and integrates it into Joules

Flow Chart: Two-Phase Routing Pipeline

OFFLINE: TRAIN & CALIBRATE



deploy trained classifier + calibrated (τ_1 , τ_2)

ONLINE: PER-PROMPT ROUTING



Low-confidence prompts escalate to a larger tier to avoid failed answers.

Algorithms

Algorithm 1: ExtractFeatures(q)

```
tokens ← split(q)
n_tokens, n_chars ← counts
avg_word_len, punct_density
flags ← def, class, import,
        return, curl, SELECT
h ← 10-d surface vector
e ← MiniLM.encode(q) # 384-d
return concat(h, e) # 394-d
```

Algorithm 2: RouteQuery(q)

```
x ← ExtractFeatures(q)
[p1,p2,p3] ← M.predict_proba(x)
if p1 ≥ τ1 and p1 ≥ p2 and p1 ≥ p3:
    return Tier 1 # 1B Q8_0
elif (p1+p2) ≥ τ2 and p2 ≥ p3:
    return Tier 2 # 3B Q4_K_M
else:
    return Tier 3 # 8B Q4_K_M
```

Default path goes to Tier 3, so uncertainty biases routing upward.

Implementation

Tier 1

Lightweight

MODEL

Llama-3.2-1B-Instruct, Gemma 3 1B,
Qwen3-1.7B

PRECISION · VRAM

Q8_0 (8-bit) · ~1.3 GB

HANDLES

Sentiment, intent, short facts

Tier 2

Intermediate

MODEL

Llama-3.2-3B-Instruct, Phi-3.5-mini0,
Qwen3-4B

PRECISION · VRAM

Q4_K_M (4-bit) · 2.6-3.1 GB

HANDLES

Summarization, NLI, JSON

Tier 3

Heavyweight

MODEL

Llama-3.1-8B-Instruct,
Mistral-7B-v0.3, Qwen3-8B

PRECISION · VRAM

Q4_K_M (4-bit) · 5.2-5.8 GB

HANDLES

Math, reasoning, code

TOOLCHAIN llama.cpp · GGUF · ONNX Runtime (MiniLM on CPU) · pynvml (NVML power) · Logistic Regression / Random Forest router

Performance Evaluation: Setup

Platform A RTX 4060

Ada · TSMC 4N · 115 W TDP
8 GB GDDR6
Idle: 54.10 ± 0.12 W

Platform B RTX 3060 Ti

Ampere · Samsung 8nm · 200 W TDP
8 GB GDDR6
Idle: 14.76 ± 0.08 W

Energy measurement

- NVML via pynvml at 20 Hz (50 ms)
- Trapezoidal integration gives E_{total}
- $E_{\text{net}} = \max(0, E_{\text{total}} - P_{\text{idle}} \cdot \Delta t)$

Benchmark: 600 prompts (200 per level)

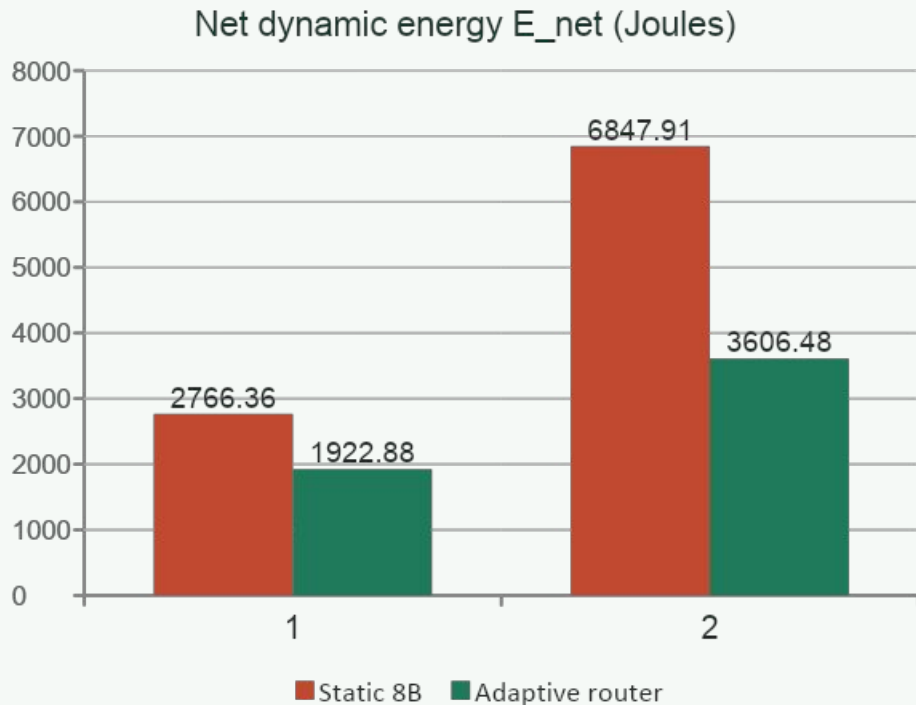
Low: SST-2 · SQuAD v2.0

Medium: CNN/DailyMail · MNLI

High: GSM8K · HumanEval

METRICS Energy: E_{total} , E_{net} , J/token, η · Quality: task score, $\text{QPR} = \text{Score}(\text{router}) / \text{Score}(\text{static 8B})$ · Latency: TTFT, end-to-end, router overhead

Results: Energy, Quality & Latency



Platform A · RTX 4060

−30.5%

net energy

90.3%

QPR

−23.9%

total energy

−20.7%

end-to-end latency

Platform B · RTX 3060 Ti

−47.3%

net energy

89.3%

QPR

−43.9%

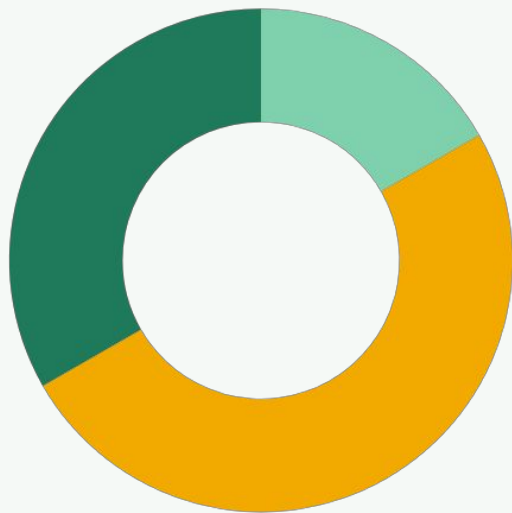
total energy

−22.7%

end-to-end latency

Results: Task-Level Breakdown

Routing decisions (12 queries)



■ Tier 1 (1B): 16.7%

■ Tier 2 (3B): 50.0%

■ Tier 3 (8B): 33.3%

Suite	Tier	Energy Δ (A)	Energy Δ (B)	Quality static $\rightarrow$ router
SST-2	1B	-65.2%	-27.0%	1.00 $\rightarrow$ 1.00
SQuAD v2.0	3B	-32.5%	-62.5%	1.00 $\rightarrow$ 1.00
CNN/DailyMail	3B	-42.8%	-28.1%	0.18 $\rightarrow$ 0.18
MNLI	3B	-72.0%	+16.4%†	0.50 $\rightarrow$ 0.50
GSM8K	8B	+4.0%†	-52.5%	1.00 $\rightarrow$ 1.00
HumanEval	8B	-21.6%	-73.7%	1.00 $\rightarrow$ 0.50

† Computed from Table 5 values. Energy = E_{net} ; A = RTX 4060, B = RTX 3060 Ti.

Results: Hardware Scaling & Overhead

MICROARCHITECTURAL SCALING η

2.475 $\times$ $\rightarrow$ **1.876 $\times$**

static 8B

adaptive router

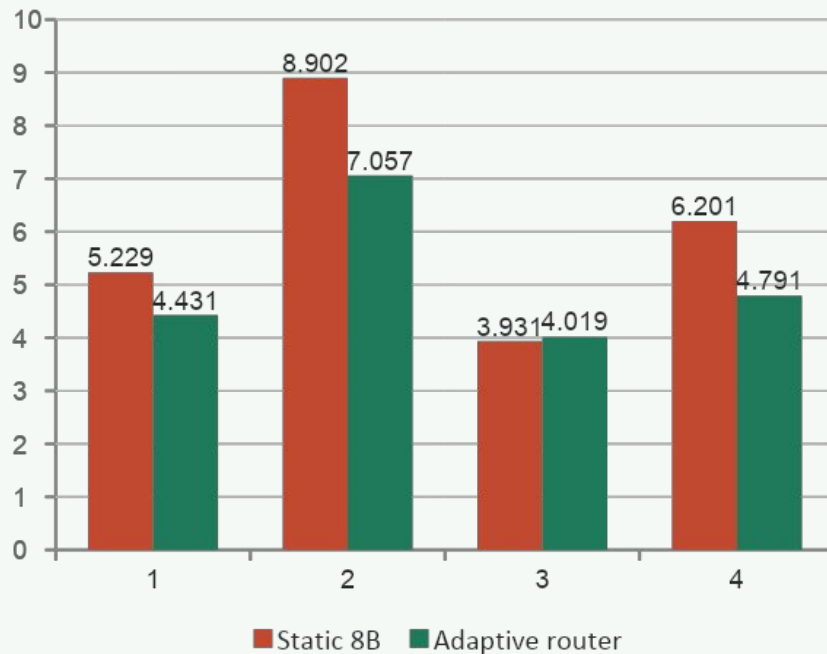
*Routing narrows the gap between GPU generations;
hardware and software gains are complementary.*

ROUTER OVERHEAD (HOST CPU)

23.43 ms · 21.88 ms

Platform A · Platform B

Serving latency (seconds)



Limitations & Future Works

Limitations

- Task quality drops ~10% (QPR 89-90%)
- Platform B J/token worse: 13.19 vs 3.29
- Router overhead ~22 ms, above the sub-5 ms target
- Small evaluation: 12 routed queries in the tier study
- Some tasks gain nothing (MNLI on B, GSM8K on A)

Future Works

- Improve router accuracy to close the quality gap
- Larger evaluation sets and more task types
- More GPUs, including data-center hardware
- Faster router and more model tiers

Conclusion

1

Routing cuts energy

−30.5% (RTX 4060) and −47.3% (RTX 3060 Ti) net dynamic energy vs. static 8B.

2

Quality mostly preserved

QPR of 90.3% and 89.3% relative to the always-8B baseline.

3

Faster, and complementary to hardware

End-to-end latency −20.7% and −22.7%; gains stack on top of newer GPU silicon.

Complexity-aware routing is a practical path to greener LLM serving.

References

- [1] Chen et al., FrugalGPT, TMLR 2024
- [2] Dettmers et al., LLM.int8(), NeurIPS 2022
- [3] Ding et al., Hybrid LLM, ICLR 2024
- [4] Fernandez et al., Energy considerations of LLM inference and efficiency optimizations, ACL 2025
- [5] Frantar et al., OPTQ, ICLR 2023
- [6] Ji & Jiang, Systematic review of electricity demand for LLMs, RSER 2025
- [7] Khan et al., Optimizing LLMs: metrics, energy efficiency, case study, IEEE CAI 2025
- [8] Li et al., Making AI less thirsty, Commun. ACM 68(7), 2025
- [9] Lin et al., AWQ, MLSys 2024
- [10] Luccioni et al., Carbon footprint of BLOOM, JMLR 24(253), 2023
- [11] Niu et al., TokenPowerBench, AAAI 2026
- [12] Ong et al., RouteLLM, ICLR 2025
- [13] Samsi et al., From words to watts, IEEE HPEC 2023
- [14] Schwartz et al., Green AI, Commun. ACM 63(12), 2020
- [15] Strubell et al., Energy and policy considerations for deep learning in NLP, ACL 2019
- [16] Wilkins et al., Hybrid heterogeneous clusters can lower LLM inference energy, e-Energy 2024
- [17] Xiao et al., SmoothQuant, ICML 2023

Thank You

Questions?

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